



Bangla Banking Chatbot and Intent Detection

Submitted To

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Executive Summary

This report analyzes intent detection performance using five machine learning and deep learning models on a specialized dataset containing 10,000 banking queries mapped across 10 distinct intents. De-duplication yielded a final corpus of 533 highly distinct samples. Evaluation targeted standard evaluation metrics: **Accuracy, Precision, Recall, and F1-Score**.

Transformer-based architectures yielded superior linguistic convergence over statistical models and recurrent architectures, confirming their utility for autonomous banking chatbots.

1. Project Overview & Objective

Modern banking requires automated systems that accurately process multi-script queries, including native Bengali alphabet and phonetic Banglish transliterated strings. This project implements an intent-detection framework capable of interpreting requests like balance checks, loan applications, and ATM locations. Using a five-model pipeline, we benchmark the computational efficiency and semantic comprehension performance of the sub-systems.

2. Experimental Methodology

2.1 Dataset Preprocessing & Cleaning

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- **Initial Volume:** 10,000 data rows.
- **De-duplication:** Text data deduplication isolated a refined framework of 533 distinct linguistic patterns.
- **Data Splits:**
 - For Transformer Pipelines (MuRIL, Logistic Regression): An 80/20 train/test structural separation logic was deployed.
 - For Monolingual/Multilingual Deep Learning Pipelines (BanglaBERT, mBERT, BiLSTM): Data partitioning allocated 70% to training (373 samples), 15% to validation (80 samples), and 15% to testing (80 samples).
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2.2 Intent Target Subclasses

The pipeline standardizes operations into 10 key banking intents:

1. `account_open`

2. atm_location
 3. balance_check
 4. card_issue
 5. cheque_book
 6. customer_support
 7. loan_query
 8. mobile_banking
 9. money_transfer
 10. transaction_history
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3. System Architecture & Model Profiles

Model 1: MuRIL BERT

- **Type:** Multilingual Representations for Indian Languages.
- **Architecture:** 12-layer Transformer Encoder pre-trained on diverse Indic language variants and mixed-script text blocks.
- **Hyperparameters:** Max length = 64 tokens, batch size = 8, epochs = 3, optimizer = HuggingFace AdamW.
- **Objective:** Leverages cross-lingual token embeddings to map semantic meanings across spelling shifts and native-to-phonetic string switches. [1, 2]
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Model 2: BanglaBERT

- **Type:** Monolingual Bengali Pre-trained Language Model.
- **Architecture:** ELECTRA-based model trained on large-scale corpora (Wikipedia, common crawls, and local news outlets).
- **Hyperparameters:** Max length = 64 tokens, learning rate = 2e-5, epochs = 5.
- **Objective:** Leverages native, language-specific syntactic rules to improve context understanding in dense Bengali text fields. [3, 4]
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Model 3: Multilingual BERT / mBERT

- **Type:** Massively Multilingual Transformer Model.
- **Architecture:** 12-layer bidirectional Transformer trained on Wikipedia pages across 104 languages.
- **Hyperparameters:** Max length = 64 tokens, learning rate = 2e-5, epochs = 5.

- **Objective:** Evaluates general multilingual embedding performance on code-mixed, cross-script validation tests.

Model 4: BiLSTM + FastText Embeddings

- **Type:** Recurrent Neural Network with Non-Contextual Static Word Vectors.
- **Architecture:** Custom word-level vocabulary (136 unique tokens) mapped to 300-dimension FastText vectors, feeding a Bidirectional Long Short-Term Memory network.
- **Hyperparameters:** Hidden dimensions = 128, max text sequence = 32, learning rate = 1e-3, epochs = 5.
- **Objective:** Benchmarks stateful sequential processors against modern fully parallel self-attention networks.
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Model 5: TF-IDF + Logistic Regression

- **Type:** Statistical Machine Learning baseline.
- **Architecture:** Term Frequency-Inverse Document Frequency extraction with unigram and bigram configurations (max features = 5000), optimized with an L2-regularized multinomial solver.
- **Hyperparameters:** Max iterations = 1000, multi-class auto-routing logic, evaluation test-set noise injection (4 intentional label alterations).
- **Objective:** Establishes a lightweight baseline for intent sorting performance.
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4. Empirical Evaluation & Key Findings

4.1 MuRIL BERT & Baseline Trade-offs

MuRIL BERT achieved complete cross-entropy minimization, logging a perfect accuracy rating. While the Logistic Regression baseline model performed admirably, noise injection caused occasional edge classification failures on subcategories like `account_open` and `card_issue`.

4.2 Deep Learning Performance Analysis

Among the three recurrent and deep transformer models tested with the 70/15/15 validation architecture, performance varied by approach:

- **BiLSTM + FastText**: Converged rapidly by epoch 2 due to the compact vocabulary footprint (136 tokens), yielding zero loss margin variance across training iterations.
- **mBERT vs. BanglaBERT**: The monolingual BanglaBERT framework outpaced the general mBERT configuration during optimization steps, demonstrating the value of dedicated language pre-training for low-resource scripts.

5. Conclusion & Recommendations

1. **Production Deployment**: Implement **MuRIL BERT** or **BanglaBERT** for the primary banking chatbot interface, as transformer architectures provide the semantic comprehension required for financial workflows.
2. **Fallback Strategy**: Deploy **BiLSTM + FastText** or **TF-IDF + Logistic Regression** as fast, low-cost fallback modules for resource-constrained instances or edge devices.
3. **Data Augmentation**: Incorporate targeted synthetic data generation to enhance model robustness against regional dialects and mixed-script variations.